

Grade 7
Unit 10
Lesson 6 Practice

Name _____
Date _____
Period _____

Use the given information to answer questions 1-2.

Lisa solved the equation $-2x - 3 = 5$. Her work is shown.

$$\begin{aligned} -2x - 3 + 3 &= 5 + 3 \\ -2x &= 8 \\ -2x \div -2 &= 8 \div -2 \\ x &= 4 \end{aligned}$$

1. Use substitution to verify if the answer is correct.
2. If correct, describe the properties of equality applied. If incorrect, describe the error(s) made.

Use the given information to answer questions 3-4.

Zoey solved the equation $3m + 5 = 8$. Her work is shown.

$$\begin{aligned} 3m + 5 - 5 &= 8 - 5 \\ 3m &= -3 \\ 3m \div 3 &= -3 \div 3 \\ m &= -1 \end{aligned}$$

3. Use substitution to verify if the answer is correct.
4. If correct, describe the properties of equality applied. If incorrect, describe the error(s) made.
5. A taxi charges \$1.75 plus a fee of \$0.75 per mile traveled. The cost of your trip was \$14.50. The equation $\$1.75 + \$0.75x = \$14.50$ can be used to represent the situation. Solve. Show your work.
6. Explain what the solution means in the context of question 5.



7. Rachael and George go ziplining. It costs a flat rate of \$75 and an additional \$5.50 for each line you cross. It cost \$124.50 per person to complete the course. The equation $\$75 + \$5.50z = \$124.50$ can be used to represent the situation. Solve. Show your work.
8. Explain what the solution means in the context of question 7.
9. Solve $9.2 = 6.1 + \frac{h}{6}$ using the properties of equality. Use substitution to verify your solution.
10. Solve $\frac{t-50}{4} = 400$ using the properties of equality. Use substitution to verify your solution is correct.